

Credit for Thin-File Customers

Credit and decision-making using alternative signals

Purple Rockets Hackathon · Team 2



AGENDA

What we'll cover

01 Context — the thin-file friction problem

02 Diagnostic — what the data shows

03 Test design — hypothesis and experiment

04 Results — early model read

05 Definition of success

06 Outcomes, trade-offs and risks

07 Next steps

CONTEXT / PROBLEM

Thin-file applicants hit friction the bureau-hit segment doesn't

One in three applications arrives with no bureau file to score — and today's process treats that as a manual problem, not a data one.

34.3%

of all applications are thin-file (178,458 of 520,000)

54% vs 19%

hit manual review — thin-file vs. bureau-hit

32h vs 3h

median decision latency, thin-file vs. bureau-hit

The friction compounds — every extra document requested raises abandonment, from 0% at zero documents to 40% at ten — and the segment still clears only **12.1%** of applications, against **81.3%** for bureau-hit. It isn't only an external-data gap either: **107,768 of these 178,458 thin-file applicants (60.4%) already bank with Nu.** We hold their deposits, balances and salary patterns — the bureau's silence doesn't mean we know nothing about them.

The bottom line: we are rejecting existing customers we already know well. Fixing that isn't just fairer to the customer — it's cross-sell revenue and a deeper relationship Nubank is currently leaving on the table.

Diagnostic: The Numbers

DIAGNOSTIC — HEADLINE NUMBERS

520K applications, one in three thin-file

520K

applications in the 12-month pilot window

34.3%

are thin-file — no bureau history

57.5%

overall approval rate
(299,205 approved)

2.59%

overall 12-month delinquency (approved book)

NORTH STAR Thin-file approval rate, while keeping expected loss constant.

DIAGNOSTIC — EXISTING NU CUSTOMERS

63% of applicants already bank with Nu

520,000 total applications

327,789 existing Nu customers 63.0% of base

199,004 approved 60.7% approval

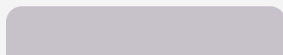
128,785 declined
actionable segment

Delinquency on the approved group holds at **2.61%**. The **128,785 declined existing customers** are the eligible population this test is built for — the segment where alternative signals can most directly change the outcome.

DIAGNOSTIC — THIN-FILE APPROVAL

Thin-file approval sits far below bureau-hit — this is the gap we target

12.1%



Thin-file approval rate

81.3%



Bureau-hit approval rate

69.2 points of gap — the opportunity this project targets.

DIAGNOSTIC — DATA PREP

Base cleaning, at a glance

Joins

application_id and customer_id join the 5 Parquet tables (applications, decisions, behavior features, underwriting ops, performance) — respecting the two granularity levels: one row per application vs. one row per account-month.

Dedup

credit_performance rolled up to one row per account_id via MAX(default_flag_12m) across its 12-month observation window before joining to decisions.

Asymmetry

Only approved accounts carry performance data (has_performance) — a deliberate, realistic gap. The pilot_holdout channel is the only place we can read repayment on would-be declines.

Leakage

Post-outcome / performance-window columns are excluded from any application-time feature set, per the target-leakage blacklist in the PRD.

Nulls

bureau_score and secured_card_utilization_pct nulls are treated explicitly — null itself is meaningful (thin-file, no secured card) — not blindly imputed.

DIAGNOSTIC — OTHER RELEVANT KPIS

Two more signals shaped the diagnostic

Pilot-holdout default rate

5.53% vs 2.42% policy

Our only clean window into true thin-file risk — the policy channel never sees these applicants. Among existing customers specifically: 5.74% (pilot) vs. 2.45% (policy).

Thin-file funnel (178,458 applications)

- **45.6%** auto path — 2.6h mean, 12.5% approved
- **44.5%** manual review — 63.9h mean, 11.7% approved
- **9.9%** abandoned before decision — 68.0h mean, 11.8% approved

Friction compounds through the funnel: manual review and abandonment both run ~25x slower than the auto path, without materially raising approval odds.

Test Design

TEST DESIGN

Testing whether behavioral variables predict creditworthiness

HYPOTHESIS

Behavioral variables — such as payroll-like recurring deposits — predict better acceptance and lower delinquency, and are usable even without a bureau score, which is exactly the thin-file gap. We test this with a **machine learning model** (Random Forest) trained on existing-customer behavior.

POPULATION

Trained on **181,151** existing customers (bureau-hit, approved, policy channel, cleaned base). Evaluated on the **9,681**-person pilot-holdout (higher-risk profiles the policy would have declined). Actionable segment: **107,768** thin-file existing customers.

MODEL VARIABLES (12, FROM APP_BEHAVIOR_FEATURES + APPLICATION DATA) — RANKED BY IMPORTANCE

1ST · 26.5% **payroll_deposit_flag**

2ND · 23.0% **secured_card_utilization_pct**

3RD · 14.4% **deposits_90d_count**

balance_avg_90d_mxn

sessions_30d

avg_session_minutes

p2p_inbound_90d_count

bill_payments_90d_count

days_since_last_login

declared_income_mxn

age

months_with_nu

Early signal: among thin-file applicants, **approved** customers show a payroll-flag rate of **49.7%** vs. **20.1%** for **declined**. Full model & notebook: [Databricks notebook](#)
→ · Live dashboard: [QuickSight](#) →

RESULTS (MODEL)

MACHINE LEARNING MODEL — EVALUATED ON PILOT-HOLDOUT

Early model read

A Random Forest classifier scores every existing customer's default risk from 0 (safest) to 100 (riskiest). Two thresholds act automatically; the 40–70 band stays on today's manual process for now.

0.68

AUC on the pilot-holdout

5.1%

FP — good customers auto-rejected, at the 70% threshold

7.7%

FN — defaulters auto-approved, at the 40% threshold

Threshold	n	% of population
Auto-approve · score < 40	30,183	28.0%
Manual · score 40–70	71,698	66.5%
Auto-reject · score ≥ 70	5,887	5.5%

APPLYING THE MODEL TO THE 107,768-PERSON ACTIONABLE SEGMENT (THIN-FILE EXISTING CUSTOMERS)

28.0% new approval rate — up from **12.1%** todaydelinquency held at **1.9%** (pilot-holdout evaluation)**2,047h** analyst hours saved via auto-decisioning

Definition of Success

DEFINITION OF SUCCESS

What success looks like

Achieve reliable results — better, lower false positives and false negatives — in the model that identifies the behavioral variables driving a low-risk Nu customer profile.

01**A reliable model**

A model whose signal holds up when checked against the pilot-holdout — not just noise.

02**A defensible low-risk profile**

A thin-file segment identified by behavioral variables that we can point to and stand behind.

03**Less manual work, less customer friction**

Fewer hours in analyst review and fewer documents requested — cutting both human intervention time and the friction thin-file customers feel.

DESIRED OUTCOMES, TRADE-OFFS & RISKS

What we want, and what could go wrong

OUTCOMES & TRADE-OFFS

- Increase the approval rate for existing Nu customers in cross-sell by **+16pp** — about **17,000** additional approved customers.
- Keep delinquency flat or lower in the test population, month over month, vs. the 2.6% portfolio baseline.
- Scope is limited to existing Nu customers with a declined decision, for now — not net-new prospects.

RISKS

- Relevant variables not captured in available datasets could be excluded from the model.
- The model may prove unreliable once deployed — it is correlational, not causal.
- Low row count: thin-file + approved + performance is a small slice (~15K), which can limit accuracy.

Next Steps

An AI-driven model turns declined thin-file customers into safe approvals — cutting analyst hours along the way

More thin-file customers included: **28.0%** new approval rate on the 107,768 thin-file existing customers, up from 12.1%

Analyst hours saved:
2,047h

01

Design a test for prospects

Add behavioral questions to the acquisition flow so non-customers can be scored the same way we score existing customers today.

02

Include non-existing customers

Extend the safe-segment approach beyond the current Nu base once prospect signals are collected and validated.

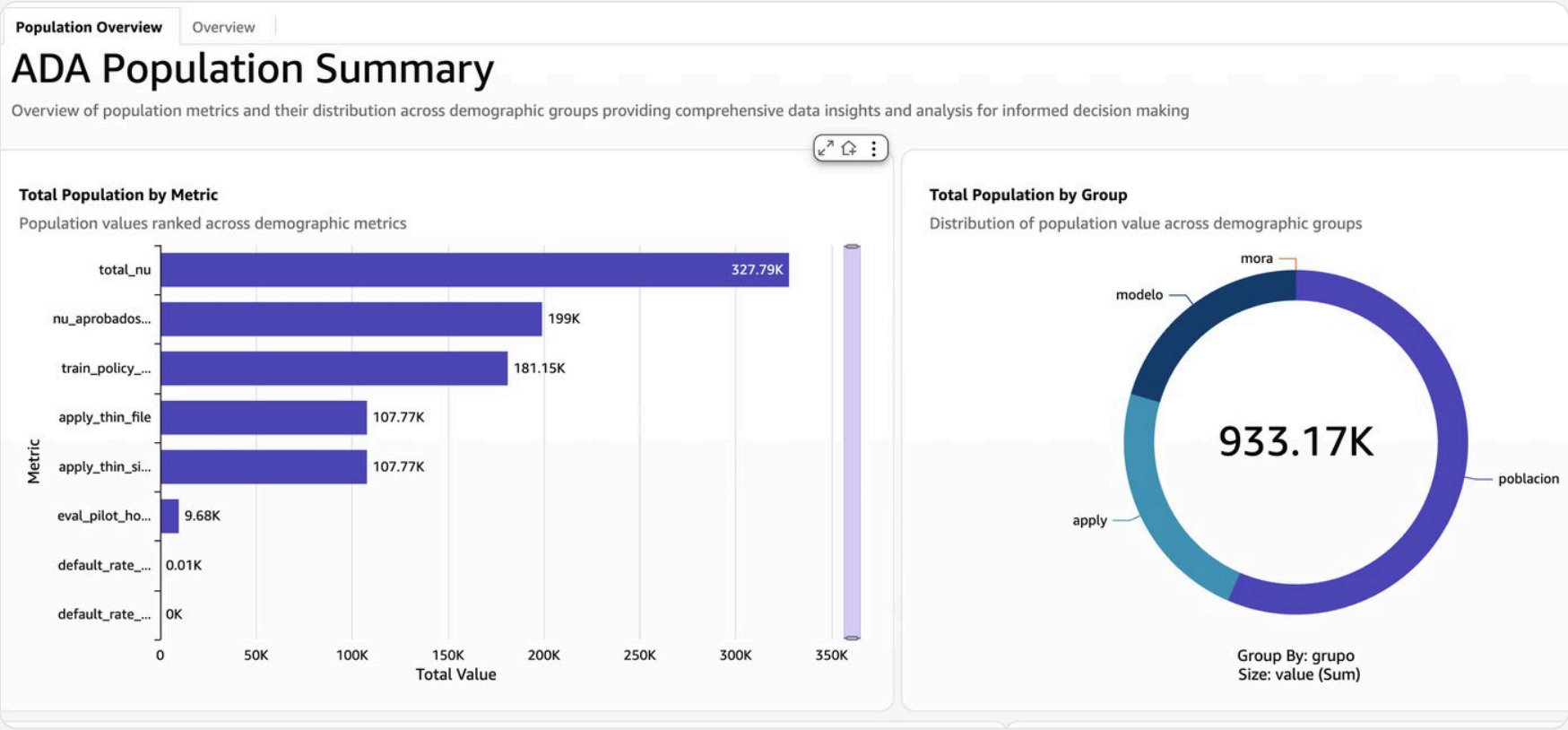
03

Roll out the new policy to more customers

Expand the safe segment gradually, monitoring approval uplift and delinquency month over month against the 2.6% baseline.

Annex

ADA Population Summary



ANNEX — QUICKSIGHT DASHBOARD

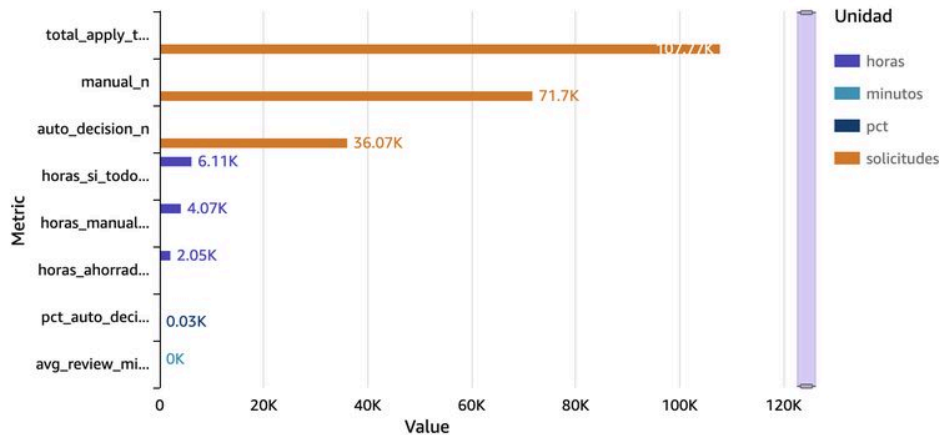
Operations, Population & Risk Bands Overview

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Combined overview showing operational savings, ADA population summary, and risk band metrics including client distribution, delinquency, and classification errors.

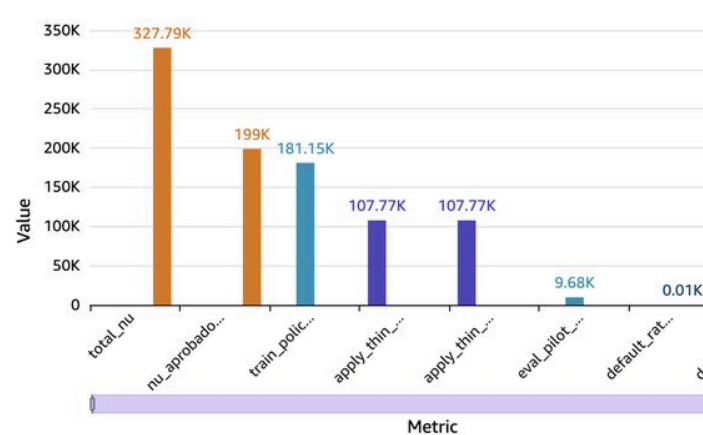
Total Operational Savings by Metric

Savings values broken down by unit of measure



ADA Population Metrics by Group

Total values across metrics segmented by group category



Operational savings (hours), population metrics by group, and the risk-band split feeding the approve/manual/reject decision. Live dashboard: [https://us-east-](https://us-east-1.quicksight.aws.amazon.com/sn/account/nu-gs-prod/dashboards/252b5fdd-d2aa-4cfb-ac70-7af40ef8ad89)

[1.quicksight.aws.amazon.com/sn/account/nu-gs-prod/dashboards/252b5fdd-d2aa-4cfb-ac70-7af40ef8ad89](https://us-east-1.quicksight.aws.amazon.com/sn/account/nu-gs-prod/dashboards/252b5fdd-d2aa-4cfb-ac70-7af40ef8ad89)